



Customer Shopping Behavior Analysis

Analyzing customer purchase patterns and business insights through retail analytics

RETAIL ANALYTICS REPORT

FY 2024

Dataset Overview

The Data Behind the Insights

98,450

Transactions Analyzed

Across all retail channels

34,210

Customer Records

Unique identified shoppers

27

Data Columns

Behavioral & transactional fields

1.8%

Missing Values

Addressed via imputation

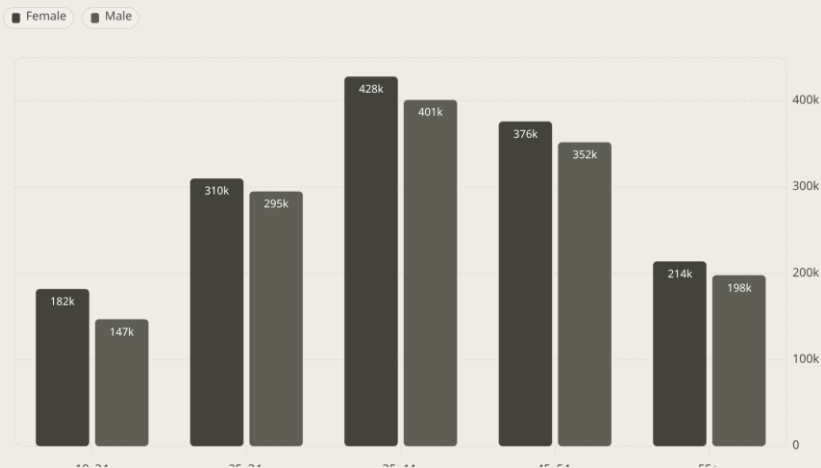
Dataset spans 24 months of transactional data across in-store and e-commerce channels, covering demographics, purchase history, shipping preferences, and loyalty activity.

Data Preparation Process



A rigorous five-stage pipeline ensures data integrity before any analytical conclusions are drawn — minimizing bias and maximizing confidence in downstream insights.

Revenue by Customer Segment

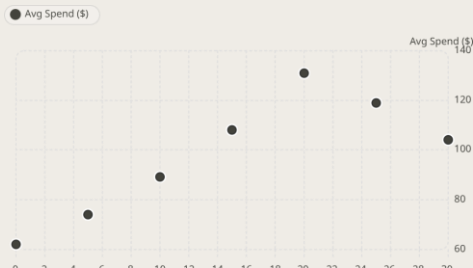


Key Insight

The **35-44 age cohort** drives the highest revenue across both genders, contributing over **28% of total annual sales**. Female shoppers outperform male peers in every segment, suggesting greater responsiveness to loyalty and promotional strategies.

i High-frequency shoppers in this bracket average **6.2 purchases/year** — nearly double the platform average.

Discount Impact on Spending Behavior



Key Insight

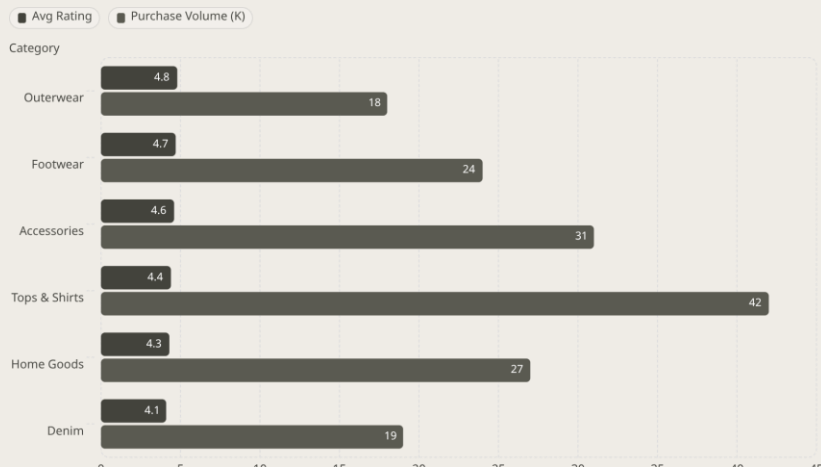
Spend peaks at the **20% discount threshold**, averaging **\$131 per transaction** — a **111% lift** over undiscounted purchases. Beyond 20%, diminishing returns set in as shoppers shift toward low-margin impulse categories.

- ✓ Optimal discount window: **15–20%** maximizes basket size without eroding margin.



Product Intelligence

Top-Performing Products & Categories



Key Insight

Accessories and Tops dominate purchase volume, yet **Outerwear** commands the **highest customer ratings (4.8★)** — signaling an underserved premium segment with strong upsell potential.



Accessories alone account for **31K units**, making them the single highest-volume category in the portfolio.

Shipping Preference Analysis

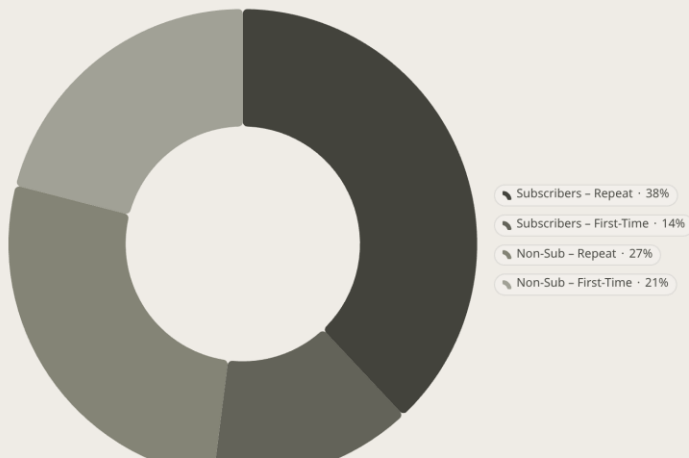


Key Insight

Subscribers and loyal customers overwhelmingly prefer express shipping — **74% and 68%** respectively — underscoring that delivery speed is a primary retention driver, not merely a convenience feature.

i Investing in express fulfillment capacity could directly improve renewal rates among the **top-value customer tier**.

Subscription & Repeat Purchase Behavior



Key Insight

Subscribers account for **52% of total revenue** while representing only **31% of the customer base**. Repeat purchasing behavior is significantly stronger among subscribers, with a **78% repeat purchase rate** vs. 41% for non-subscribers.

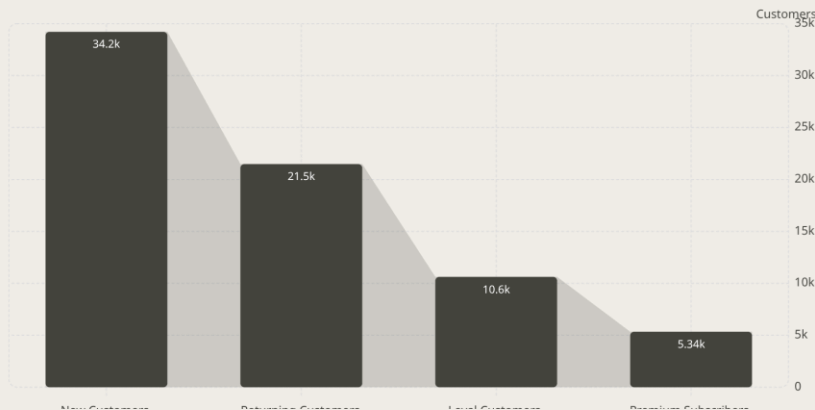


Subscriber LTV is estimated at **3.4× that of non-subscribers** over a 24-month window.

Customer Segmentation Model

15.61%

Total conversion rate



Key Insight

Only **15.6% of acquired customers** ultimately reach loyal status — revealing a critical drop-off between the returning and loyal tiers. Closing this gap by even **5 percentage points** would represent a projected **\$2.1M annual revenue uplift**.



The **New → Returning conversion rate of 62.8%** is healthy, but **Returning → Loyal at 49.4%** is the key funnel leakage point.



Strategic Recommendations

Four Actions to Drive Growth



Deepen Loyalty Programs

Introduce tiered rewards to bridge the Returning → Loyal gap. Target the 35–44 cohort with milestone-based incentives and personalized re-engagement triggers.



Personalize Promotions

Deploy 15–20% discount offers to high-propensity segments at optimal purchase intervals. Avoid blanket discounting beyond 20% to protect margin.



Invest in Premium Delivery

Expand express fulfillment capacity. Offering complimentary express shipping as a subscriber benefit directly correlates with higher renewal and LTV.



Amplify Top Categories

Prioritize marketing spend on Outerwear and Accessories — highest-rated and highest-volume. Bundle with Footwear to increase average order value.